
Handwriting Author Identification

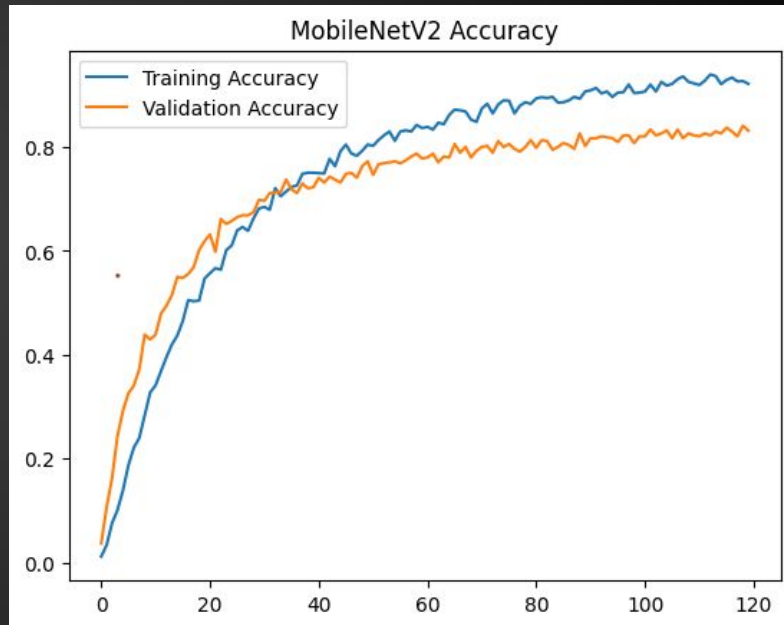


Zainab Siddiqui
Data Capstone II

Summary



- ✦ **Switched focus from EfficientNetB1 to MobileNetV2**
 - Smaller
 - Lowered learning rate ($1e-3 \rightarrow 1e-5$)
 - Unfreezing layers
 - 384x384
 - Accuracy $\rightarrow$ top-3 accuracy
- ✦ **Augmentations**
 - Zoom, then crop, then downsize
 - Train set
 - Did not work
- ✦ **Alt. train/val/test split based on prompt**
 - Random split
 - Consistent split

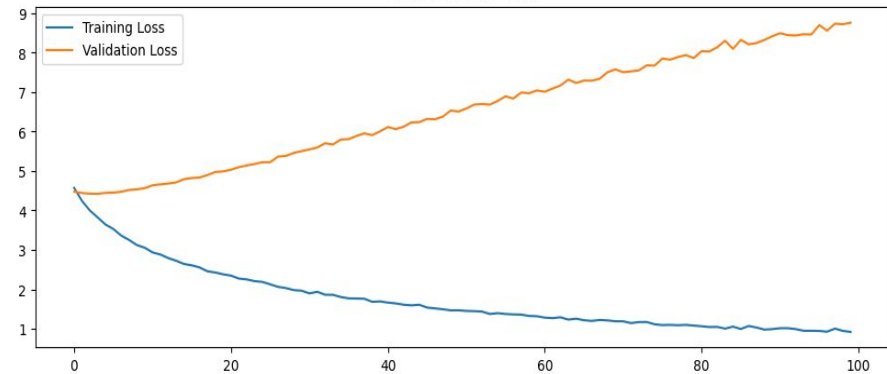




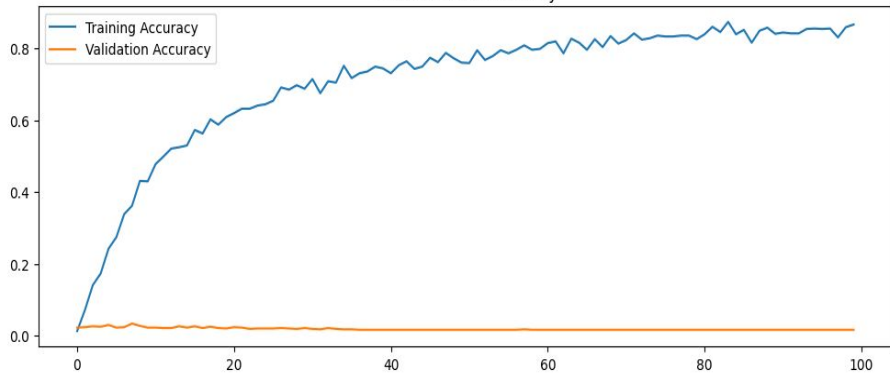
MobileNetV2

EfficientNet vs. MobileNet

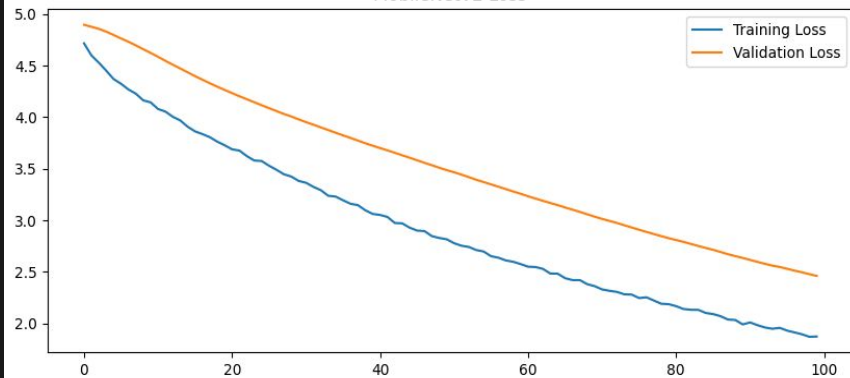
EfficientNetB1 Loss



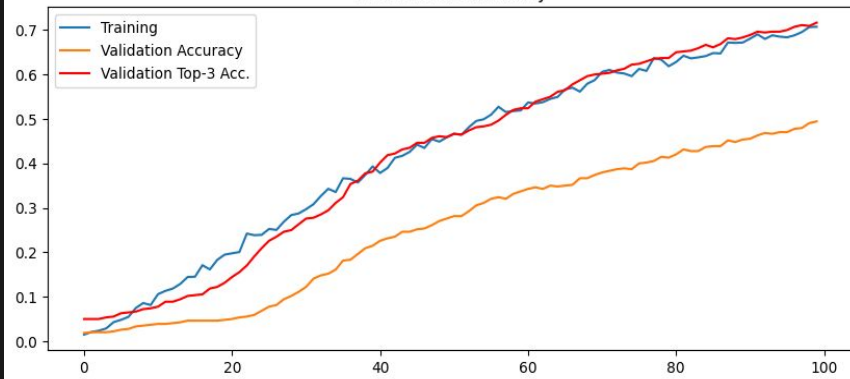
EfficientNetB1 Accuracy



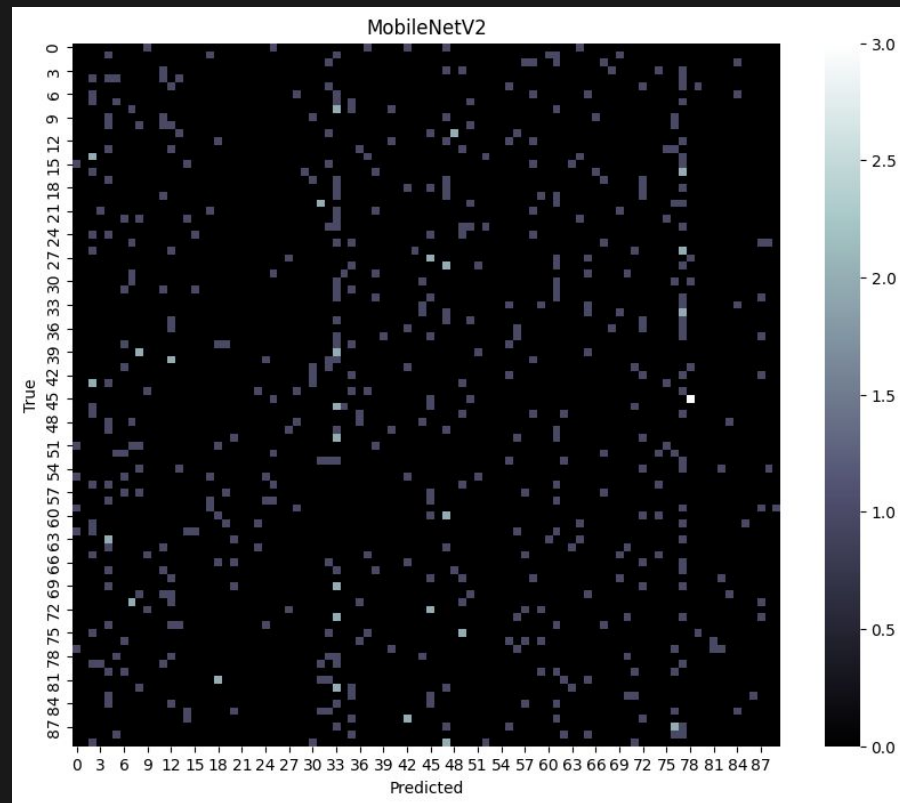
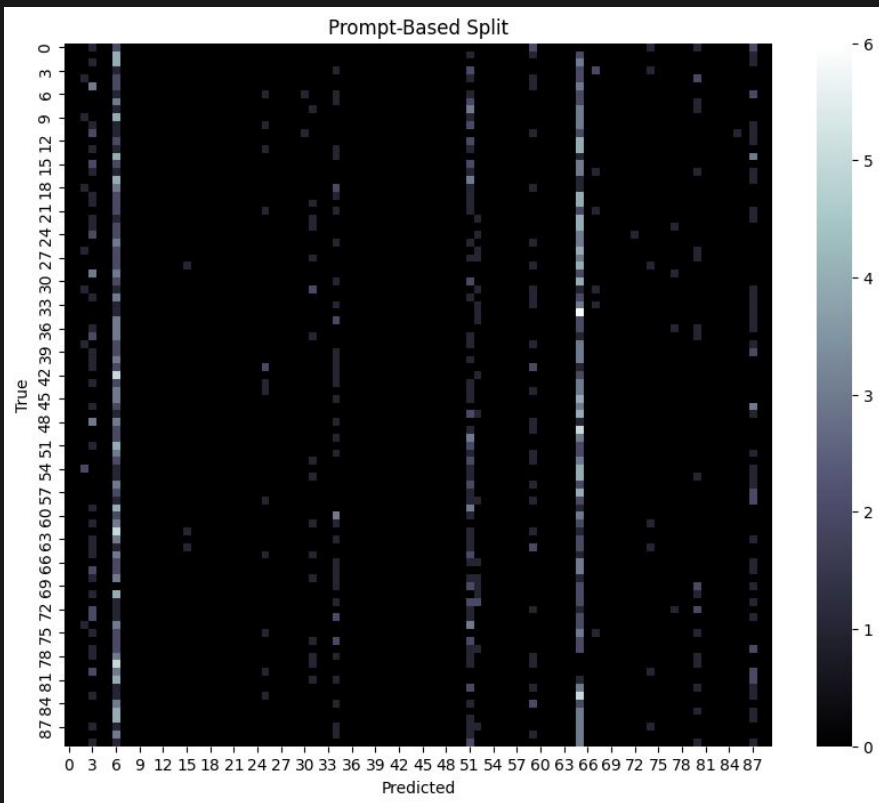
MobileNetV2 Loss



MobileNetV2 Accuracy



: Problem with MobileNetV2





33/33/33 Split

London Letter

Our London business is good, but Vienna and Berlin are quiet. Mr. O. Lloyd has gone to Switzerland and I hope for good news. He will be there for a week at 1469 Zernoth Street and then goes to Zurich and Rome and will join Colonel Parry and arrive at Athens, Greece, November 27th or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Tuttle Tuesday. Dr. L. McQuaid and Robert Vreger, Esq., left on the 'Y.V.' Express tonight.

Proverb

The early bird may get the worm, but the second mouse gets the cheese.

Wizard of Oz

Within a short time she was walking briskly toward the Emerald City, her silver shoes tinkling merrily on the hard, yellow road bed. The sun shone bright and the birds sang sweet and Dorothy did not feel nearly as bad as you might think a little girl ~~who~~ would who had been suddenly whisked away from her own country and set down in the midst of a strange land.

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope for good news. He will be there for a week at 1496 Zermott Street and then goes to Turin and Rome and will join Colonel Parry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Fuller Tuesday, Dr. L. McQuaid and Robert Unger, Esq., left on the 'Y. X.' Express tonight.

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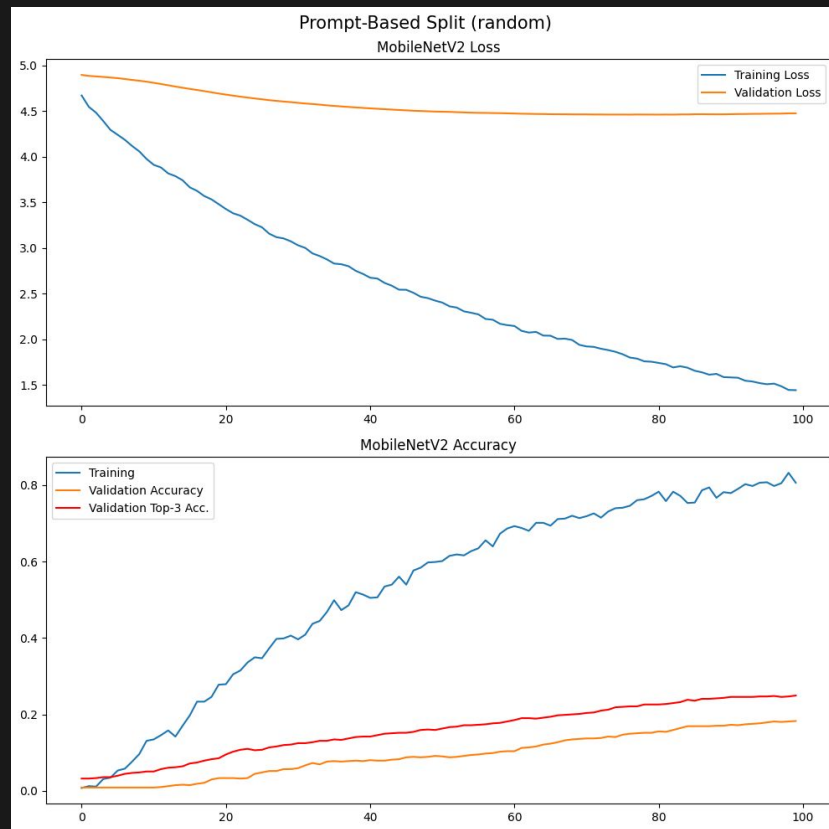
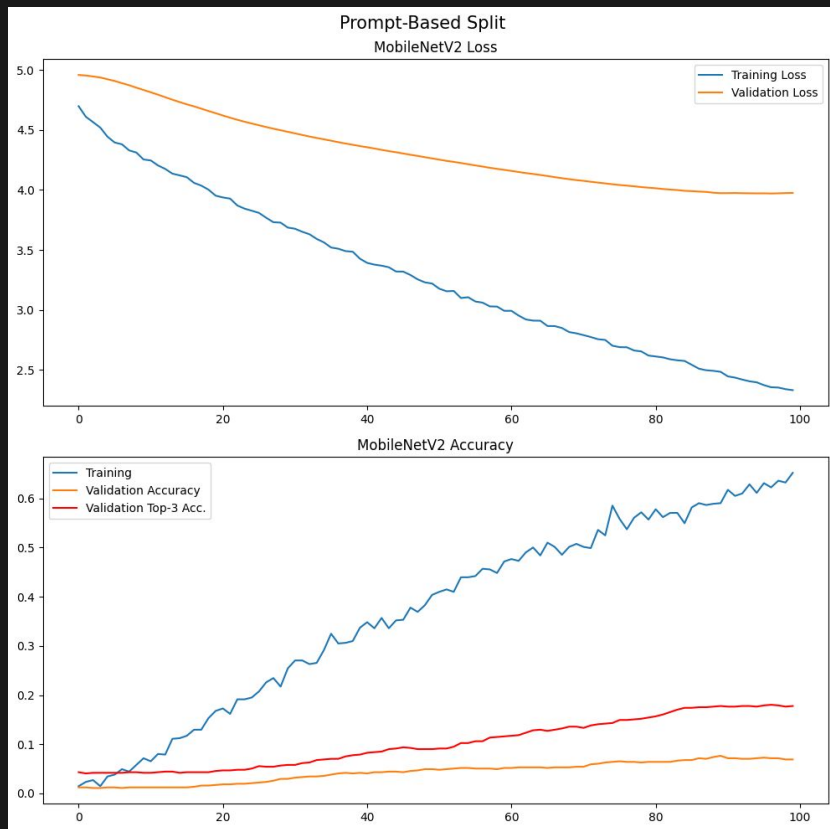
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Consistent vs. Random





Augmentations

Previous

November 27 or

L. addressed King
James E. Fuller
+ Unger, Esq.,
1.

Switzerland and I hope
there for a week a
then goes to Turin and
Rome and arrive at Athens
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Current

82

Blvd. 3580. We expect Charles E. Fuller Tuesday. Dr. L.
Robert Unger, Esq., left on the 'Y.X.' Express tonight.

16

mainly on the hard, yellow roadbed.
shone bright and the birds sang sweet
nothing did not feel nearly as bad as I
think a little girl would who had
whisked away from her own
and set down in the midst of a
land.

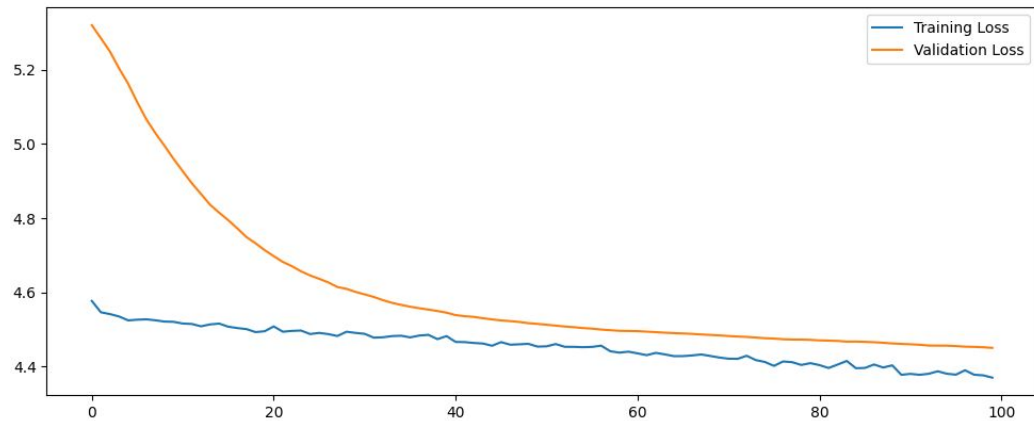
18

Berlin are quiet. Mr. D. Lloyd has gone to
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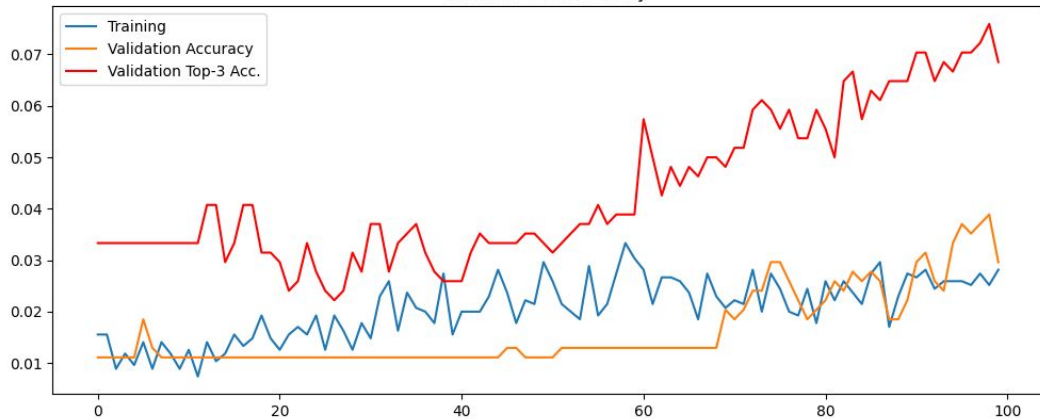
29

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Augmentations
MobileNetV2 Loss



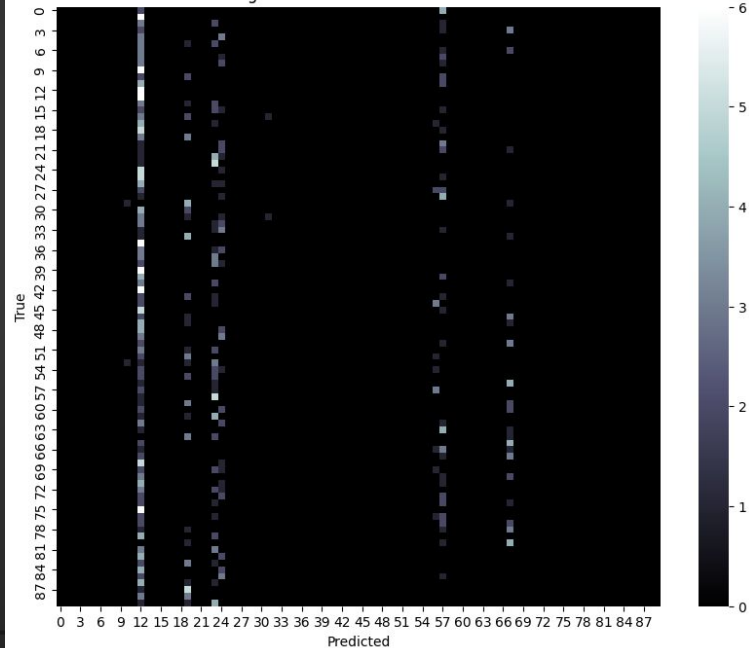
MobileNetV2 Accuracy



Augmented Data



Augmentations Confusion Matrix





Results

	Current Baseline	33/33/33 (same)	33/33/33 (random)	Augmented
Train Accuracy	71%	64%	83%	2.8%
Val. Top-3 Acc.	72%	18%	25%	6.8%
Test Accuracy	46%	6.2%	10%	3.5%
Test Top-3 Acc.	68%	13%	24%	8.9%



Future Work



- ✦ **Debug and fine-tune augmentations**
- ✦ **Retry EfficientNet baseline**
- ✦ **Custom CNN**
- ✦ **Full dataset (~475 classes)**

- ✦ **Deployment**
- ✦ **Explainable AI**
 - **GRAD-CAM(?)**
 - **PCA + UMAP projection**



Questions?

